

Defect-Driven Lattice Dynamics and Finite-Temperature Stability in $B2$ NiTi

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Abstract

Nickel–titanium (NiTi) is a shape-memory alloy characterized by a reversible martensitic transformation between the high-temperature $B2$ austenite phase and the low-temperature $B19'$ martensite phase. Lattice dynamics plays a central role in this transformation, making the assessment of phase stability at finite temperature essential.

In this work, we investigate the impact of point defects—specifically vacancies and antisites—on the lattice dynamics and free energy of the $B2$ phase at finite temperature. Short-range disorder induced by these defects is quantified through Warren–Cowley parameters up to the fourth nearest neighbors. Molecular dynamics simulations are combined with Harmonic Ensemble Lattice Dynamics (HELD) to extract effective harmonic force constants within the Born–von Kármán framework, including interactions up to the fifth nearest neighbors. To enable systematic and scalable analysis, HELD is integrated into the Utilities To Execute Pipelines (UTEP) framework.

Our results confirm the dynamical stability of the $B2$ phase at finite temperature, consistent with previous studies. Increasing temperature leads to phonon stiffening at the M high-symmetry point across all configurations. Vacancies produce only minor modifications to the phonon spectra, whereas antisite defects induce a noticeable shift of phonon modes toward higher energies and promote hybridization between acoustic and optical branches at the X point. Both types of defects increase the internal energy and the entropic contribution, suggesting a thermodynamic stabilization at elevated temperatures.